



Statistics and Analytics



MARKS DISTRIBUTION FOR EACH CHAPTER		
Sl. No.	Chapters	Marks
1	Statistical Data Collection and Types	3
2	Summarization of Data	6
3	Measure of Location and Dispersion	5
4	Introduction to Python Programming	6
	Total	20

Statistical Data Collection and Types

1

1.1 INTRODUCTION:

Statistics stands for the numerical information from which certain conclusions can be drawn. In this sense the word statistics is used as “figure or numerical information”. When man started counting, comparing and keeping records in one or other way, the subject ‘statistics’ seems to have originated. Sir Ronald A. Fisher is considered as father of Statistics. The Indian scholar Prashanth Chandra Mahalanobis called as father of Indian statistics.

The word statistics seems to have been derived from the Latin word '**status**' or the Italian word '**statista**' or the German word '**statistik**' or the French word '**statistique**'. Statistics is a science of collection, representation, analysis and interpretation of numerical data.

For example:

- “Sales statistics of different products of a company for past few years are recorded for a comparative study.
- The Height of group of 6 boys in cms are: 158, 164, 178, 168, 156, 174.

In these sentences the term statistics refers to numerical statements.

Main Divisions of Statistics (Branches)

Statistics have two branches, namely **Statistical Methods** and **Applied Statistics**.

Statistical Methods are also called mathematical statistics or theory of statistics. It deals with the procedure of statistical analysis of numerical data. It is a tool for decision making.

Applied Statistics is the application of statistical methods to concrete situations like Agriculture, Industry, Population, Medicines etc. Some branches of Applied Statistics are,

Some branches of Applied statistics

1. Biometry (Bio statistics which deals with problems in Biology)
2. Econometrics (Econometrics is a tool to prove or disprove the theories of economics)
3. Demography (Study human population)
4. Statistical quality control (S.Q.C.) (control quality of manufactured goods)
5. Actuarial science (Statistics in the field of insurance)
6. Stylometry (Statistics in literature)
7. Psychometry (Statistics in psychology)

Function of statistics

1. It presents the facts in a definite form.
2. It simplifies and condenses the size of the data.
3. It facilitates comparison.
4. It enriches our knowledge and widens our experience.
5. It helps in formulating policies.

Limitations of statistics

1. It does not deal with individuals
2. It does not deal with qualitative data.
3. Statistical laws are true only on an average.
4. Only experts can make the best possible use of Statistics.
5. Statistics is liable to be misused.

Some basic concepts of Statistics

Units or Individuals

The objects whose characteristics are studied in any statistical survey are called **units** or **individuals**.

can provide more natural feedback than individual interviews and are easier to organize than experiments or large-scale surveys.

1.6 DATA CLEANING

Data cleaning, also known as data cleansing or data scrubbing, is the process of identifying and correcting or removing errors, inconsistencies, and inaccuracies in a dataset. It is an essential step in the data preparation process to ensure the accuracy and reliability of the data.

Key considerations for data cleaning:

Remove duplicate entries: Identify and remove any duplicate records in the dataset. Duplicates can occur due to data entry errors, system glitches, or merging of

datasets. By eliminating duplicates, you avoid skewing the analysis and ensure each observation is unique.

Check for data consistency: Examine the relationships between different variables in the dataset to ensure consistency.

Conduct data quality checks: Perform various data quality checks, such as checking for data accuracy, completeness, and reliability.

Document data cleaning procedures: Keep a detailed record of the steps taken during the data cleaning process. Document the changes made, the reasons behind them, and any assumptions or transformations applied. This documentation helps ensure transparency and reproducibility.

PRACTICE MCQ's

- The data classification which is based on variables such as production, cost, height and weight is considered as**
 - qualitative classification
 - quantitative classification
 - open end classification
 - time series classification
- The discrete variables and continuous variables are two types of**
 - open end classification
 - time series classification
 - qualitative classification
 - quantitative classification
- The classification of data on the geographical basis is also called**
 - reflected classification
 - populated classification
 - sampling classification
 - spatial classification
- The set of raw data arranged in ascending or descending order is called**
 - ordered array
 - nominal array
 - ordinal array
 - interval array
- The classification on the basis of time order is called as**
 - disclosed classification
 - chronological classification
 - external classification
 - internal classification
- If the classification of collected data is based on characteristics such as religion, education and caste, then this is considered as**
 - open end classification
 - time series classification
 - qualitative classification
 - quantitative classification
- The process of arranging data on the basis of certain properties in classes or groups is classified as**
 - classification of data
 - reflection of data
 - sample of population
 - sample observation
- The father of statistics**
 - R.A. Fisher
 - A.L. Bowley
 - Horace Secrist
 - Boddington

Summarization of Data

In our day to day life, recording of information is very crucial. A piece of information or representation of facts or ideas which can be further processed is known as data. The weather forecast, maintenance of records, dates, time everything is related to the collection of data. The collection, presentation, analysis, organization and interpretation of observations of data is known as statistics or data summarization. Using statistics, predictions about the nature of data can be made based on the previous data is to be studied and observed.

The statistical data which is collected can be represented by various methods such as tables, bar graphs, pie charts, histograms, frequency polygons etc.

In this unit representation and analysis of data is discussed such as:

- Descriptive statistics
- Grouped data
- Frequency distribution Table
- Relative frequency Table
- Bar graph
- Pie chart
- Line graph
- Frequency polygon
- Frequency curve
- Relative frequency polygon
- Histogram
- Box plot
- Leaf-stem plot

2.1 DESCRIPTIVE STATISTICS

Descriptive Statistics, as the name suggests, describes data. It is a method to collect, organize, summarize, display and analyze sample data taken from a population.

Descriptive Statistics is not based on probability theory. It paves the way to understand and visualize data better.

2.1.1 Grouped Data

Data can be defined as groups of information that represent the qualitative or quantitative attributes of a variable or set of variables, which is the same as saying that data can be any set of information that describes a given entity. Data in statistics can be classified into grouped data and ungrouped data.

Any data that you first gather is ungrouped data. Ungrouped data is data in the raw. An example of ungrouped data is any list of numbers that you can think of. Grouped data is data that has been organized into groups known as classes. Grouped data has been 'classified' and thus some level of data analysis has taken place, which means that the data is no longer raw.

A data class is group of data which is related by some user defined property. For example, if you were collecting the ages of the people you met as you walked down the street, you could group them into classes as those in their teens, twenties, thirties, forties and so on. Each of those groups is called a class.

Each of those classes is of a certain width and this referred to as the Class Interval or Class Size. This class interval is very important when it comes to drawing Histograms and Frequency diagrams. All the classes may have the same size or they may have different classes sizes depending on how you group your data. The class interval is always a whole number.

2.2 FREQUENCY DISTRIBUTION TABLE

The frequency (f) of a particular observation is the number of times the observation occurs in the data. The distribution of a variable is the pattern of frequencies of the observation. Frequency distributions are portrayed as frequency tables, histograms or polygons.

Measure of Location and Dispersion

3

Charts are frequently very useful to visually represent data, they are inconvenient for the simple reason that they are difficult to display and can not be remembered. It is frequently useful to reduce data to a couple of numbers that are easy to remember, easy to communicate, yet capture the essence of the data they represent.

In this unit statistical data measure of location and dispersion is studied mainly

- Determination of central tendencies Mean, Mode and Median for the data.
- Determination of absolute measures of dispersion for data like range, quartile deviation, mean deviation, standard deviation and variance.
- Skewness and kurtosis graphs and interpretation of results.

3.1 DETERMINATION OF CENTRAL TENDENCIES

We consider a random variable X and a data set $S = \{x_1, x_2, \dots, x_n\}$ of size n which contains possible values of X . The data set can represent either the population being studied or a sample drawn from the population.

We seek a single measure (i.e., statistic) which somehow represents the center of the entire data set S .

Central tendency : It is the property of concentration of the values around a central value in a data.

Average (Measure of central tendency)

It is a single value that represents the entire mass of data.

Objectives

1. To get a single value to represent the characteristics of the entire mass of data.
2. To facilitate comparison.

Essentials of a good central tendency

1. It should be easy to understand and simple to calculate
2. It should be based on all the observations.
3. It should be well defined by a mathematical formula.
4. It should not be affected by abnormal extreme values
5. It should be capable of further algebraic treatment.
6. It should possess sampling stability
7. It should be easily calculated for an open-end distribution.

Types of averages

1. Arithmetic mean
2. Median
3. Mode
4. Geometric mean
5. Harmonic mean

3.1.1 Arithmetic mean (Mean)

Arithmetic mean of a set of observations is their sum divided by the total number of observations. It is denoted by $\bar{x}$. That is

$$\bar{x} = \frac{\text{sum of the observations}}{\text{Total number of observations}}$$

Case (1) : Ungrouped data (raw data)

If $x_1, x_2, \dots, x_n$ are ' n ' observations in a data then

$$\bar{x} = \frac{x_1 + x_2 + \dots + x_n}{n} = \frac{\sum x}{n}$$

Case (2) : Grouped data

$$\bar{x} = \frac{\sum fx}{N}$$

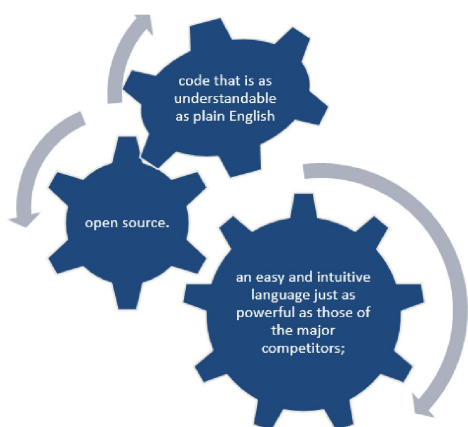
Introduction to Python Programming

4

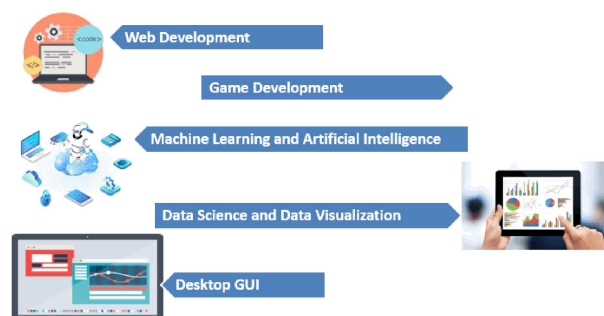
4.1 INTRODUCTION TO PYTHON

- It was created by Guido van Rossum, and first released on February 20, 1991.
- Python is a widely-used, interpreted, object-oriented, and high-level programming language with dynamic semantics, used for general-purpose programming.
- It has a simple syntax similar to the English language and it allows developers to write programs with fewer lines than other programming languages.
- It can be used on a server to create web applications, create workflows, connect to database systems, read and modify files, handle big data and perform complex mathematics. It can implement object oriented features, exception handling
- *Object-oriented programming (OOP)* is a programming paradigm that allows you to keep similar data together, and give the ability to modify the data

Importance of python programming



Real World Applications



4.2 PYTHON SYNTAX

Python syntax can be executed by writing directly in the Command Line

```
>>> print("Hello, World!")  
Hello, World!
```

creating a python file on the server, using the .py file extension, and running it in the Command Line:

```
C:\Users\Your Name>python myfile.py
```

4.3 PYTHON COMMENTS

Comments can be used to explain Python code.

Comments can be used to make the code more readable.

Comments can be used to prevent execution when testing code.

Types of Comments in Python

there are 2types of comments.

- 1.Single-Line Comments
- 2.Multi-Line Comments